

Hellenic Complex Systems Laboratory

Inferences about the Difference between Two Proportions

Technical Report XVIII

Theodora Chatzimichail and Aristides T. Hatjimihail
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Inferences about the Difference between Two Proportions

Theodora Chatzimichail ^a and Aristides T. Hatjimihail (Aristeidis T. Chatzimichail) ^a

^a Hellenic Complex Systems Laboratory

Search Terms: proportions, difference between proportions, confidence interval, inference, critical ratio statistic, statistical significance, statistics

Short Description of the Demonstration

This Demonstration explores statistical inference for the difference between two population proportions associated with the presence of a condition or trait. It computes the statistical significance and confidence interval for the difference between the two proportions and plots the corresponding confidence limits as functions of the selected confidence level. The calculations are performed for different counts of individuals with and without the condition, and for different confidence levels used in interval estimation.

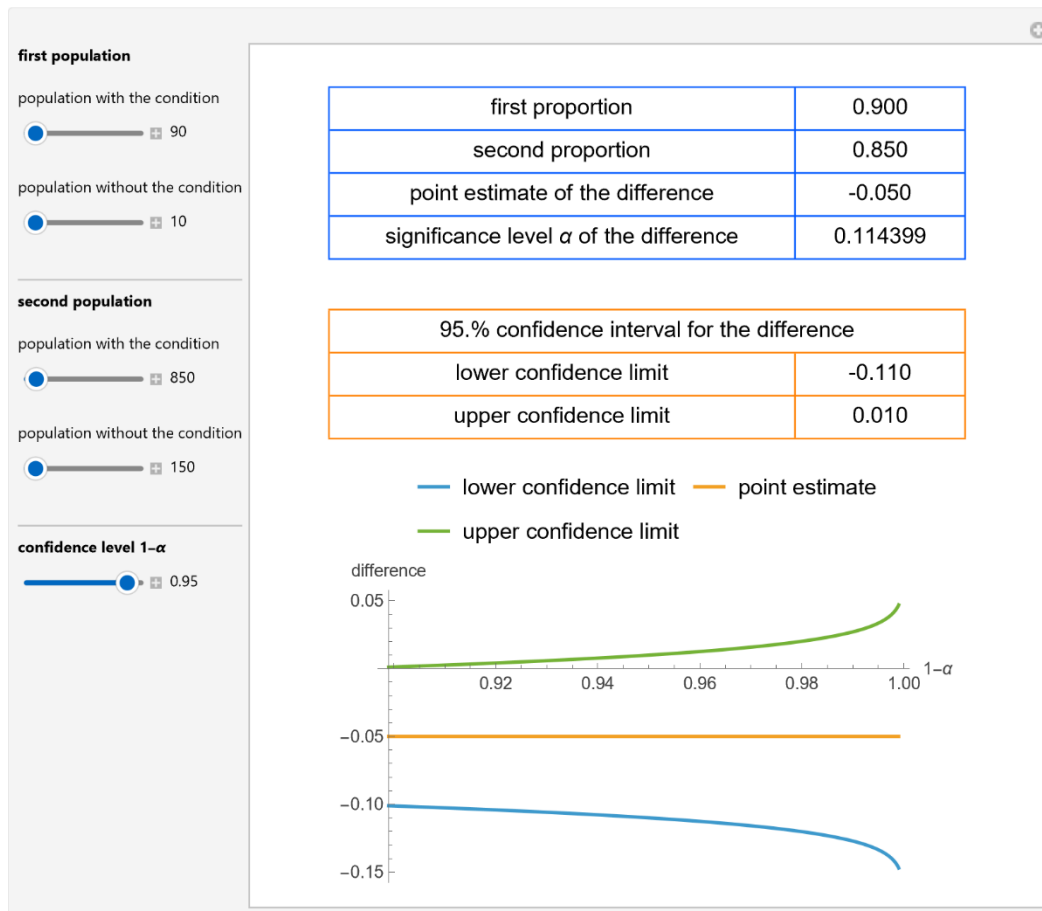


Figure 1: Point estimate and 95% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus confidence level for the difference, with the settings shown at the left.

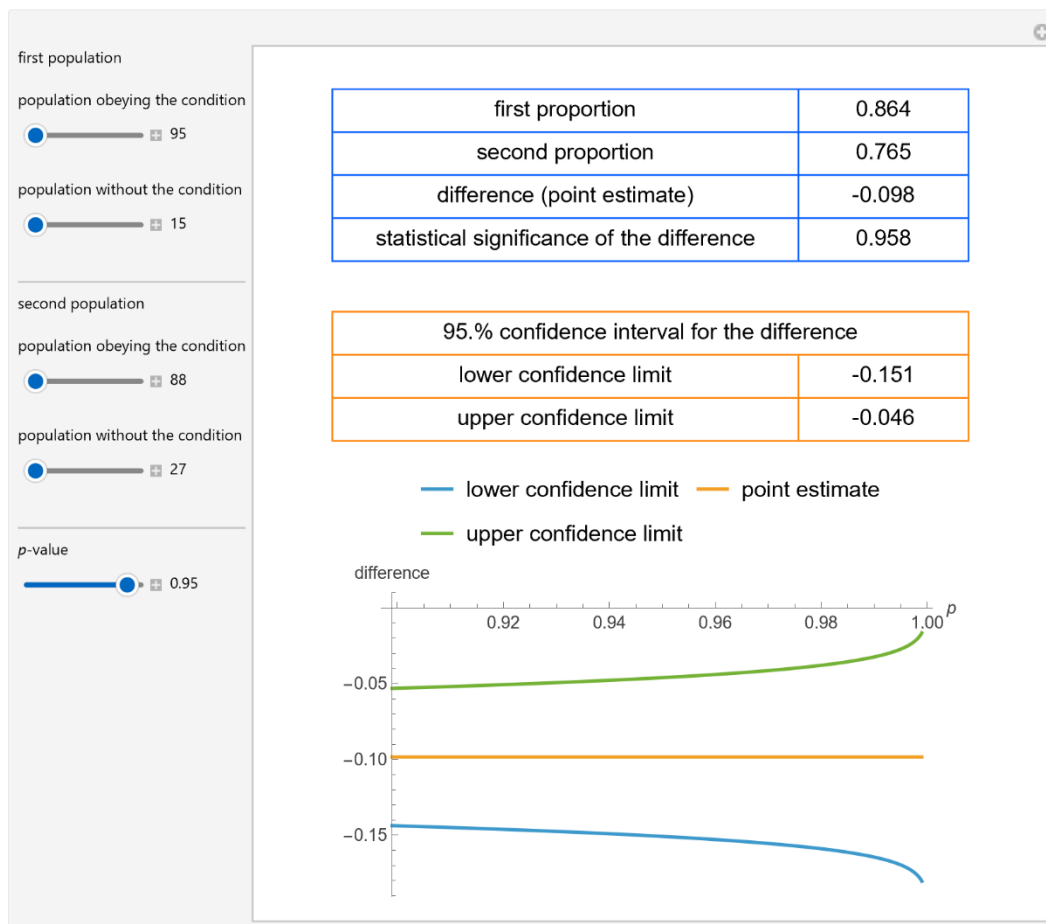


Figure 2: Point estimate and 99% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus confidence level for the difference, with the settings shown at the left.

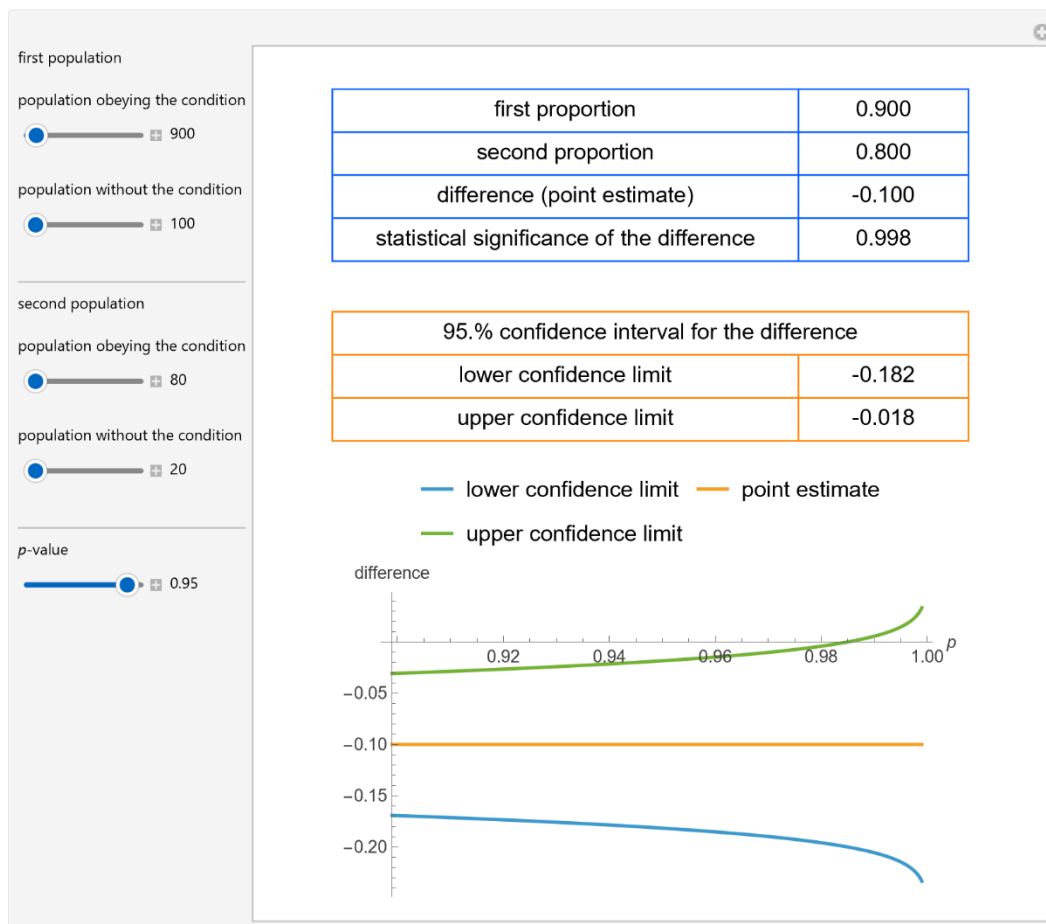


Figure 3: Point estimate and 95% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus confidence level for the difference, with the settings shown at the left.

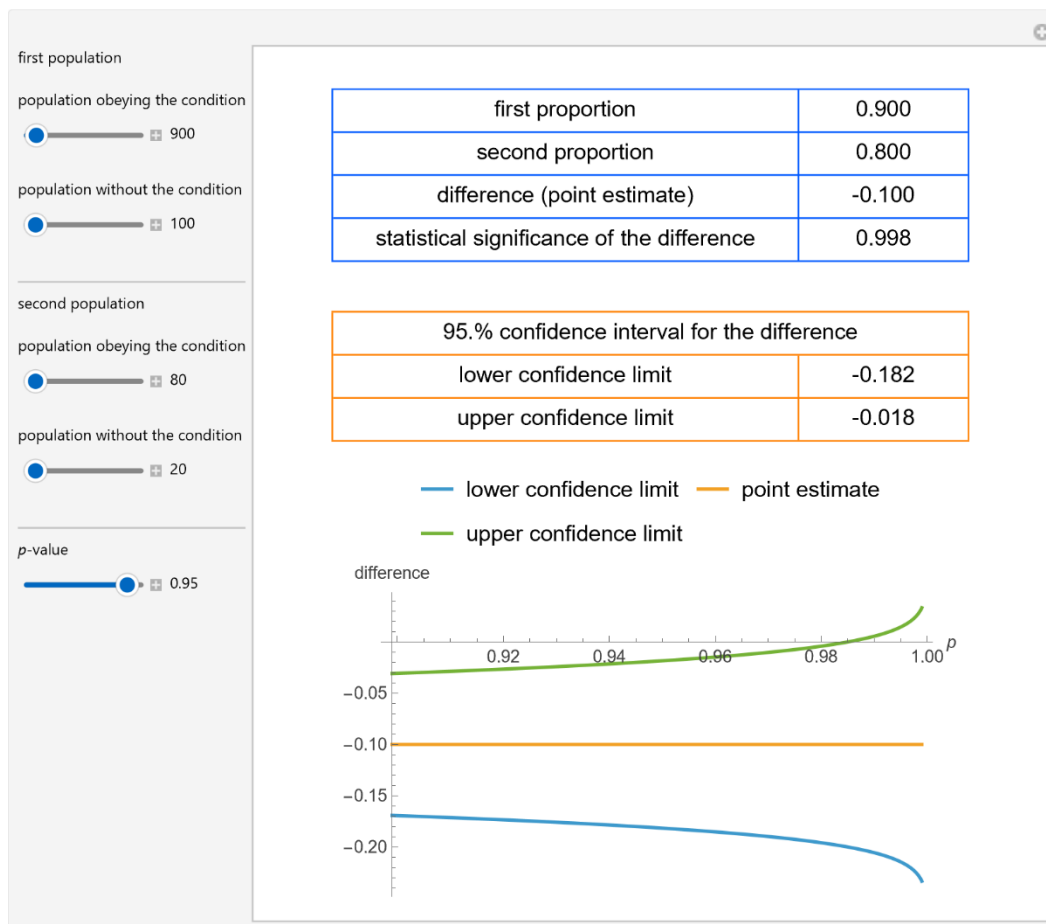


Figure 4: Point estimate and 95% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus confidence level for the difference, with the settings shown at the left.

Details

The significance level α of the difference between the two proportions is assessed by means of the z-score (or critical ratio statistic) [1]. The confidence interval for the difference between the proportions at a confidence level $1-\alpha$ is calculated as in [2].

References

[1] J. L. Fleiss, B. Levin and M. C. Paik. Methods for Generating a Fourfold Table. *Statistical Methods for Rates and Proportions*, 3rd ed., Hoboken, NJ: J. Wiley, 2003 pp. 51–52.

[2] J. L. Fleiss, B. Levin and M. C. Paik. A Simple Confidence Interval for the Difference between Two Independent Proportions. *Statistical Methods for Rates and Proportions*, 3rd ed., Hoboken, NJ: J. Wiley, 2003 pp. 60–61.

Demonstration

Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:

<https://www.hcsl.com/Tools/Demonstrations/InferencesAboutTheDifferenceBetweenTwoProportions.nb>

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

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<https://demonstrations.wolfram.com/InferencesAboutTheDifferenceBetweenTwoProportions/>

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